

Overview Of Our Services

Advanced Molecular Detection Southeast Region

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ELC Updates

- BP3 Project D. AMD Updates – budgets, milestones, project approaches due in CAMP in **35 days!!**
- Internally your deadlines may be different
 - Check in with your ELC officer!
- Apply for BTL In-person Training travel for 2 Bioinformaticians in 1b.
 - This is a 2-day workshop in Athens, GA
 - Topic is TBD – more info on next slide

BTL Trainings

- Online Training
 - 15-week series, split up into 5 3-week cycles
 - Week 1 – Concept Seminar
 - Week 2 – Landmark Case Study
 - Week 3 – Applied Practical
 - Audience is all public health employees – analysts, laboratorians, bioinformaticians, epidemiologists, and leadership.
- In-person Training
 - Tentatively in Athens, GA at UGA campus
 - Please apply for 2 Bioinformaticians to travel for 2 days
 - Topic is TBD but may focus on a leadership workshop for supporting expanded use of molecular data for public health policy, response and surveillance

Github

- StaPH-B GitHub link ([StaPH-B Github](#))
- StaPH-B Southeast Region GitHub link ([StaPH-B Southeast Region](#))
- StaPH-B YouTube ([StaPH-B – YouTube](#))
- BPHL-Molecular GitHub link ([BPHL-Molecular · GitHub](#))

Pipeline Trainings

Bacterial Genomics Pipelines	
Training	Summary
Amelia	Targeted NGS of <i>Mycoplasma genitalium</i> ; QC, amplicon stats, SNPs, AA variation
Aquascope	Environmental & waterborne bacterial genome analysis; QC, assembly, characterization
BigBacter	Large-scale bacterial WGS; QC, assembly, typing, AMR detection
FLAQ-AMR	Standard bacterial assembly pipeline for AMR detection & taxonomy
Flisochar	Bacterial isolate characterization using hybrid assembly
Flisochar_ontlr	Web-deployed FLISOCHAR for ONT long reads
Flisochar_wf	Web-deployed FLISOCHAR for ONT + PacBio long reads
Fl_cg_snp	SNP pipeline for identifying clusters of related isolates
GentySpuds	Web-deployed Enterovirus genotyping
Neissflow	WGS pipeline for Neisseria; assemblies, SNPs, resistance detection
Legionella Tools	Genomic characterization, typing, genotyping, surveillance
PHoeNIX	Bacterial pathogen genome analysis with QC, typing, plasmids, and AMR detection
Sanibel	Faster Nextflow version of FLAQ-AMR for bacterial genomes
Talbot	Pan-genome analysis + phylogenetic tree generation
PhyloPhoenix	Phylogenetic tree generation for bacterial genomes

Pipeline Trainings

Viral Pipelines	
Training	Summary
Daytona	Nextflow version of FLAQ_SC2; adds human read removal
Daytona_cl	Nextflow version of the FLAQ_SC2_Clearlabs workflow
Daytona_combine	Combines Daytona + Daytona_cl; supports Clearlabs & Illumina
Daytona_dengue	Dengue NGS analysis pipeline
Daytona_Pb	SARS-CoV-2 analysis from PacBio
Daytona_WNV	West Nile Virus genome analysis; mapping %, coverage, consensus, VADR, SNPs
Enterovirus_Assembler	De novo Enterovirus assembly from fastq
FLAQ_MPX	Monkeypox clinical specimen pipeline; consensus + variant calling
FLAQ_SC2_Clearlabs	Generates SARS-CoV-2 consensus assemblies; supports Illumina & non-Illumina
FLAQ_SC2_Meta	SARS-CoV-2 metagenomics (wastewater); runs Freyja
MIRA-NF	Respiratory virus genome assembly (Influenza, SARS-CoV-2, RSV)
PolkaPox	Poxvirus genome analysis; consensus + variant call

Pipeline Trainings

Fungal Pipelines	
Training	Summary
Pensacola	Candida auris long-read (PacBio) analysis
Mycosnp	Reference-based SNP + phylogenetic analysis for fungal pathogens

Environmental & Waterborne Pipelines	
Training	Summary
Aquascope	Environmental & waterborne bacterial genome analysis; QC, assembly, characterization

Pipeline Trainings

General Microbial/WGS/Variant/Phylogenomics	
Training	Summary
Sarek Mic	Microorganism-focused Sarek variant pipeline; SNP calling & annotation
Varpipe	General WGS variant analysis; alignment, variant calling, annotation
Mashwrapper	Rapid genome similarity estimation using Mash

Reporting & Submission Tools	
Training	Summary
SeqSender	Customized CDC SeqSender for FL BPHL; submission prep & upload
ReportGen	Automated standardized report generator (QC, tables, metadata)

HPG Trainings

Training & Developer Tools	
Training	Summary
Basic Bash	Linux fundamentals: navigation, file handling, basic scripting
VS Code Basics	Intro to VS Code for bioinformatics workflows; extensions, terminal, remote HPG use

Datacamp/APHL

Github Training

- Introduction to Git ([Link](#))
- Intermediate Git ([Link](#))
- Introduction to Github Concepts ([Link](#))

Machine Learning

- Understanding Machine Learning ([Link](#))

AWS

- AWS Concepts ([Link](#))
- AWS Cloud Technology and Services Concepts ([Link](#))



Advanced Molecular Detection

Southeast Region Bioinformatics

Questions?

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